

PROJECT 1 – EZRENTAL AUTO RENTAL POS MANAGEMENT SYSTEM DATABASE DESIGN AND IMPLEMENTATION

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PROJECT 1 – EZRENTAL AUTO RENTAL POS MANAGEMENT SYSTEM DATABASE DESIGN AND IMPLEMENTATION

EXECUTIVE SUMMARY

The EZRental Auto Rental POS Management System project was completed to design and implement a centralized enterprise database solution supporting customer identity and payment management for rental locations and digital channels. The project followed best practices in database development and Agile delivery, ensuring secure, scalable, and reliable data models aligned with EZRental's business objectives.

The database design included conceptual, logical, and initial physical modeling phases, focusing on customer, credit profile, and payment information domains. Final deliverables included the complete project documentation, conceptual ER/EER model, normalized logical schema, business and technical requirements, and Agile feature backlog. This foundation enabled future development of rental transaction features, vehicle inventory, branch operations, and online booking modules.

PROBLEM STATEMENT & OBJECTIVES

EZRental required a secure, scalable, and standardized data foundation to support customer onboarding, profile management, and payment processing across physical rental offices, corporate systems, and online customer portals. Prior to this initiative, the organization operated without a unified database architecture capable of supporting global growth, multi-channel operations, and future automation capabilities.

To address this gap, NYC-Tech Solutions was engaged to design and initiate implementation of the foundational database system for EZRental's next-generation enterprise rental platform. The scope of this phase focused on developing the core customer and payment data structures that would enable reliable identity management, payment verification, and secure account handling across all business channels.

The objective of the project was to deliver a normalized and scalable database architecture supporting EZRental's operational needs. This architecture was designed to serve employees within rental agency locations, corporate intranet users responsible for business operations and reporting, and retail customers accessing services through public online channels. The solution prepared the foundation for enterprise-wide rental workflows, including reservation processing, rental transactions, vehicle fleet management, and customer self-service functionality.

This deliverable established the first phase of EZRental's platform modernization strategy, ensuring strong data integrity, efficient customer validation, improved data quality, and readiness for future expansion across global markets including the United States, Canada, Mexico, United Kingdom, Japan, and Australia, with planned expansion into Asia, Africa, and Mediterranean regions.

PROJECT MANAGEMENT METHODOLOGY

The EZRental Auto Rental POS database project was executed using a hybrid project management methodology that combined elements of both the Waterfall Project Management Methodology and the Agile Development Methodology. This blended approach aligned with real world enterprise software delivery practices, supporting structured planning and documentation while enabling iterative development and continuous refinement throughout the project lifecycle.

The Waterfall framework established the foundation for project planning, requirements gathering, conceptual modeling, logical modeling, and physical design before development began. This ensured the effort followed a structured and traceable delivery roadmap, consistent with enterprise grade database implementation practices. Key Waterfall activities included business and technical requirement definition, conceptual ER/EER data modeling, logical model development, and planning of physical schema standards prior to build and testing phases.

Agile principles were integrated to support incremental development across four sprint cycles, allowing continuous validation, progressive refinement, and phase-by-phase delivery of design and implementation artifacts. Scrum-style sprint cycles were used to plan and execute deliverables, including documentation drafts, model integration, and database design checkpoints. Feedback loops and iterative improvements were performed at the conclusion of each sprint to ensure alignment with business objectives and technical goals.

This hybrid methodology enabled structured project execution supported by iterative development and continuous improvement. The combination of Waterfall and Agile ensured clear documentation, consistent planning, quality control, and progressive delivery aligned with the expectations of EZRental's leadership and enterprise system development standards.

DATABASE DESIGN DELIVERABLE #1A - APPLICATION BUSINESS REQUIREMENTS

- The following pages contain the ***Business Requirements captured*** by the **Business Analyst**. For simplicity, note the following legend that describes the highlighting markings in the business requirements:

The system was designed to support EZRental's core customer-facing business processes by delivering:

- Secure storage of customer identity and demographic information
- Management of multiple payment profiles per customer
- Support for fraud-prevention data validation and business rules
- Centralized storage for branch and corporate rental access
- Scalability to support future rental booking and vehicle assignment workflows
- Ability to be extended into web-based corporate and consumer portals

Business rules were developed to enforce accurate data capture, maintain referential integrity, and support enterprise reporting needs.

BUSINESS REQUIREMENTS

ABOUT US:

EZ-Car Rental is an auto rental company that rents vehicles such as cars, SUVs, minivans & cargo vans to customers. In addition, other specialized vehicles such as trucks, motorcycles, boats, mobile homes, etc. We operate in several countries with rental agency locations in the US, Canada, Mexico, UK, Japan & Australia. Within each country we own and operate rental agencies located in cities, regions and state. For example, New York City has 2 rental agencies in Manhattan, one in Brooklyn and two in Queens located at each airport. With multiple rental agencies in cities, states etc., a customer can pick up a vehicle in one location and drop it off at another.

CURRENT CHALLENGES:

Our current rental system is outdated, with a poor user-experience, inefficient (breaks often thus expensive to operate), does not meet our business requirements, and is not scalable (cannot be easily updated with new features). Another very important shortcoming of the current system is the lack of elasticity since it does not give us the flexibility to scale-up or scale-down resources during business trends and seasonal changes in the market.

We want to invest in modernizing our business with a new vehicle management system that can meet these challenges and delivers a great user-experience, meet our new business requirements, scalable, and elastic to adopt to business trends and seasonal market changes. Elasticity is very important since recently we have been faced with a new type of competition; small rental companies that are nimble and can quickly adopt to market changes thus able to provide new offerings that are appealing to customers thus affecting our profits. These smaller competitors are using new technologies that enable them to be nimble and elastic. Figurative speaking *“they are eating our lunch”*.

We look forward to your proposed architecture & implementation of this new system. Below are our business requirements.

OUR RENTAL AGENCIES:

A **rental agency** is the location where customers visit to mainly pick up and drop off rental vehicles, in addition to making reservations. Rental agencies also contain a vehicle maintenance department that manage the maintenance and repairs of the rental vehicles in the agency. In addition, a rental agency has an inventory department to track and manage the inventory of the vehicles for the specific agency. The management of vehicle inventory is owned by a global inventory team that manage the inventory for the entire fleet of cars, nevertheless, each rental agency has a local team that manage the vehicle inventory for a specific agency. In short, the inventory is managed locally within an agency and globally by the corporate global inventory team. In summary, all inventory from all our agencies need to synchronize with the global inventory system.

Rental Agencies also have a transport department with a team of drivers that transport vehicles from one rental agency location to another agency or a vehicle warehouse etc., as part of managing the daily rental business. Vehicle transportation is required since customers can pick up a car in one location and drop it off at another, therefore, the transport department will manage the pickup and delivery of vehicles from one agency to their assigned rental agencies. In addition, new vehicles are transported to their assign agencies from our central warehouses throughout a country.

A very important category of rental agencies is those located at Airport and Train Hubs. These rental agencies have heavy traffic of customers arriving and departing to their destination. **EZRental Inc.**, want to make sure our customers have the best experience, especially during peak seasons and rush hours and don't have to wait long to be able to pick up or drop off their vehicles. Therefore, we have the following business requirements that the system need to support:

- The Auto Rental System at the agencies must deliver high-performance. It needs to be fast and quick to execute transactions to minimize long lines and expedite a customer transaction asap. Customers should not be waiting long on a line; we want to expedite their transactions quickly and effectively.
- Customer Service Representatives at Rental Agencies in Airports and Train Hubs, should not have to use a slow system or must wait long for a transaction to complete. The system should be fast and provide a great user experience to our Customer Service Representative engaging with customers.

The data that we need to capture in the Auto Rental System is as follows:

BUSINESS REQUIREMENTS

OUR CUSTOMERS:

EZ-Car Rental offer their services to two types of **Customers: Corporate Customers & Retail Customers**. **Corporate Customers** are individuals whose corporation have a contract with us to use our services with special corporate rate for their employee's rental services. On the other hand, **Retail Customers** are consumers not associated with a company and engaging in personal rental.

Requirements for All Customers (Retail & Corporate Customers)

To run our business, the application must store the following customer information for both types of **customers (retail & corporate)** so this data is common to both types of customers:

- A **Customer ID** number which uniquely identifies the customer, **customer name** which is composed of: **first name, last name**.
- **Birth date, Age, Address** which includes the elements: **address line1, address line 2** (which is optional and used for apartment number, suite or any additional address information required), **city, state code** (which is the two-character code for a state in the US), **zip code & country**.
- Customer **phone number & email** (unique like all emails and required to rent).
- In addition, a driver license is required to reserve and rent a vehicle. Therefore, we need to capture the unique **driver license number** (an alpha numeric character string containing numbers & characters. Note that in the USA the format for the driver license number can be under 15 characters, but in other countries it could go up to 22 to 25 characters) **driver license expiration Date** and **driver license state**. In addition, note the following **business rule** policy regarding the business importance of the **driver license number**:

1. *The driver license number is used throughout the business to identify a customer for searching, reporting etc.*
2. *Therefore, the driver license number is the main unique ID for a customer to be identified and managed from a business perspective.*

- A very important attribute we need to capture for every customer is the **credit card**. For our credit card processing and transactions, we need to capture the following **credit card** components: **credit card number** that uniquely identifies the credit card and is a 16-character number digits. We also need to capture the **credit card owner name**, in addition, credit card processing attributes such as **credit card issuing bank code, credit card issuing bank name, credit card network company code, credit card network company name, credit card processing merchant service company code, credit card merchant service company name, credit card corporate merchant bank code, and credit card corporate merchant bank name**. Important – further details on these credit card processing attributes will be provided in sections to follow. Is important that these attributes are clearly understood to correctly design the system.
- Other attributes of credit card are **expiration date, billing address** composed of **address line1, address line 2** (which is optional and used for apartment number, suite or any additional address information required), **city, state code** (which is the two-character code for a state in the US), **zip code & country, credit card limit** (which is the maximum amount of money a customer can charge on their credit card), **credit card available credit** (which is how much you have left to spend with your credit card or unused amount within your limit). Note that we will capture the credit card limit to a maximum of \$999,999.99, since we don't expect our customers to have a credit limit of \$1 Million dollars. Finally, **activation status** (which is true if the credit card is active and can be used, or false when the credit card is not active or disabled).
- During the interview with business stakeholders provided the following **Business Rules** related to a credit card:

1. *You cannot reserve or rent one of our vehicles without a credit card.*
2. *A customer can have many credit cards they can use to pay for rental transactions.*
3. *A credit card can be owned by the one customer or co-owned by other individuals such a family member or corporate entity the customer works for. Therefore, many customers can own the same credit card and a credit card can be owned by many customers.*

BUSINESS REQUIREMENTS

OUR CUSTOMERS (CONT.):

The Customer Credit Card processing Workflow

Processing of the credit card transactions is a key part of this business and is important that we store data for each step of the credit card processing process. Therefore, the credit card processing attributes discussed in previous section need to be further analyzed and clearly understood. We will now provide the definition and detailed information on each of these credit card processing attributes: ***credit card issuing bank code, credit card issuing bank name, credit card network company code, credit card network company name, credit card processing merchant service company code, credit card merchant service company name, credit card corporate merchant bank code, and credit card corporate merchant bank name:***

- ***Credit Card Processing Merchant Service Company***– In credit card processing the merchant is the retailer where a customer purchased the goods and services and pay using a credit card. EZRental Inc., is the merchant in this scenario. The ***Credit Card Processing Merchant Service Company*** is the institution which works directly with the merchant (EZRental Inc.) to provide handle the credit card processing services and handles all the complexity of credit card processing and interactions with the other financial entities involved in the credit card processing process on behalf of the merchant (EZRental Inc.). The ***Credit Card Processing Merchant Service Company*** provides the Merchant or Business with the hardware which the customer swipes or inserts to pay for goods and services with their credit card. As part of the credit card processing cycle, the first financial institution which the ***Credit Card Processing Merchant Service Company*** interacts with is the ***Credit Card Network Company*** (which we will cover next). The ***Credit Card Processing Merchant Service Company*** ensure the merchant (EZRental Inc.) is connected to the right ***Credit Card Network Company***. We will describe the ***Credit Card Network Company*** next, nevertheless, we need to capture the following information for the ***Credit Card Processing Merchant Service Company***:

- ***Credit Card Processing Merchant Service Company Code***– In our business, we use and store a number code used to identify the ***Credit Card Processing Merchant Service Company***. This code has business meaning and appears in reports and discussed by users. Therefore, a ***Credit Card Processing Merchant Service Company*** can be identified by this code.
- ***Credit Card Processing Merchant Service Company Name***– In our business, we also use and store the name of the ***Credit Card Processing Merchant Service Company***. This name also has business meaning and appears in reports and discussed by users. Therefore, a ***Credit Card Processing Merchant Service Company*** can be identified by its name.
- Below is a listing of the values/instances of the ***Credit Card Processing Merchant Service Company Code*** and ***Credit Card Processing Merchant Service Company Name*** we use at EZRental Inc:

<i>Credit Card Processing Merchant Service Company Code</i>	<i>Credit Card Processing Merchant Service Company Name</i>
1	Stax by Fattmerchant
2	Helcim
3	Dharma Merchant Services
4	Payment Depot
5	National Processing
6	Block
7	Intuit Quickbooks
8	PayPal
9	Stripe
10	Flagship Merchant Services
11	Clover

BUSINESS REQUIREMENTS

OUR CUSTOMERS & CREDIT CARD PROCESSING (CONT.):

- **Credit Card Network Company**– In credit card processing the objectives of the **Credit Card Network Company** is to process transactions between the **Credit Card Issuing Bank** (which we will cover next) and the **Credit Card Processing Merchant Service Company**. Covered previously. The **Credit Card Network Company** act like bridges between the **Credit Card Issuing Bank** that issue credit card and the **Credit Card Processing Merchant Service Company** that handles the transaction from the merchant (EZRental Inc.). The **Credit Card Network Company** that interacts with the **Credit Card Issuing Bank** to determine whether to approve or deny the transaction. And then the **Credit Card Network Company** notifies the merchant (EZRental Inc.) if the purchase was approved or denied. The **Credit Card Network Company** is a digital infrastructure that facilitates credit card transactions and prepares the transaction for the **Credit Card Issuing Bank**. We will describe the **Credit Card Issuing Bank** next, nevertheless, we need to capture the following information for the **Credit Card Issuing Bank**:

- **Credit Card Network Company Code**– We use and store a number code to identify the **Credit Card Network Company**. This code has business meaning and appears in reports and discussed by users. Therefore, a **Credit Card Network Company** can be identified by this code.
- **Credit Card Network Company Name**– In our business, we also use and store the name of the **Credit Card Network Company**. This name also has business meaning and appears in reports and discussed by users. Therefore, a **Credit Card Network Company** can be identified by its name.
- Below is a listing of the values/instances of the **Credit Card Network Company Code** and **Credit Card Network Company Name** we use at EZRental Inc.:

Credit Card Network Company Code	Credit Card Network Company Name
1	American Express
2	Visa
3	Mastercard
4	Discover
5	Diners Club
6	Interlink
7	Star
8	Accel
9	Interac
10	Visa ReadyLink
11	Pulse
12	JCB (Japan Credit Bureau)
13	Rupay

Credit Card Issuing Bank Code	Credit Card Issuing Bank Name
1	American Express
2	Bank of America
3	Barclays
4	Capital One
5	Chase
6	Citi
7	Discover
8	Synchrony Bank
9	U.S. Bank
10	Wells Fargo

- **Credit Card Issuing Bank**– The **Credit Card Issuing Bank** is the financial or lending institution that offers the Credit Card and pays for the goods and services until the customer pays back the credit/loan. These are Banks, Lending Institutions, Credit Unions, Fintech companies, etc. These institutions issues/provides the credit for the customer. The cardholder borrows money from the credit card issuing bank each time they make a purchase, and when they pay their credit card bill, they're paying the **Credit Card Issuing Bank**. We need to capture the following information for the **Credit Card Issuing Bank**:
- **Credit Card Issuing Bank Code**– In our business, we use and store a number code used to identify the **Credit Card Issuing Bank**. This code has business meaning and appears in reports and discussed by users. Therefore, a **Credit Card Issuing Bank** can be identified by this code.
- **Credit Card Issuing Bank Name**– In our business, we also use and store the name of the **Credit Card Issuing Bank**. This name also has business meaning and appears in reports and discussed by users. Therefore, a **Credit Card Issuing Bank** can be identified by its name.
- Below is a listing of the values/instances of the **Credit Card Issuing Bank Code** and **Credit Card Issuing Bank Name** we use at EZRental Inc.:

BUSINESS REQUIREMENTS

OUR CUSTOMERS & CREDIT CARD PROCESSING (CONT.):

- **Credit Card Corporate Merchant Bank**– In credit card processing, the *Credit Card Corporate Merchant Bank* is the bank used by the merchant **EZRental Inc.**, to handle credit card processing money transactions, payments, etc., between **EZRental Inc.**, and the *Credit Card Processing Merchant Service Company* that handles the Credit Card Processing on behalf of **EZRental Inc.** In short, it is the bank that has the bank account used by **EZRental Inc.**, to handler the accounting for Credit Card Processing Transactions. **EZRental Inc.**, currently uses three banks to handle all their banking transactions. They don't expect to add many Corporate Merchant Banks throughout the next 5+ years, but business is everchanging, nevertheless, expectations is that there are no plans to add any new banks no time soon.
- We need to capture the following information for the *Credit Card Corporate Merchant Bank*:

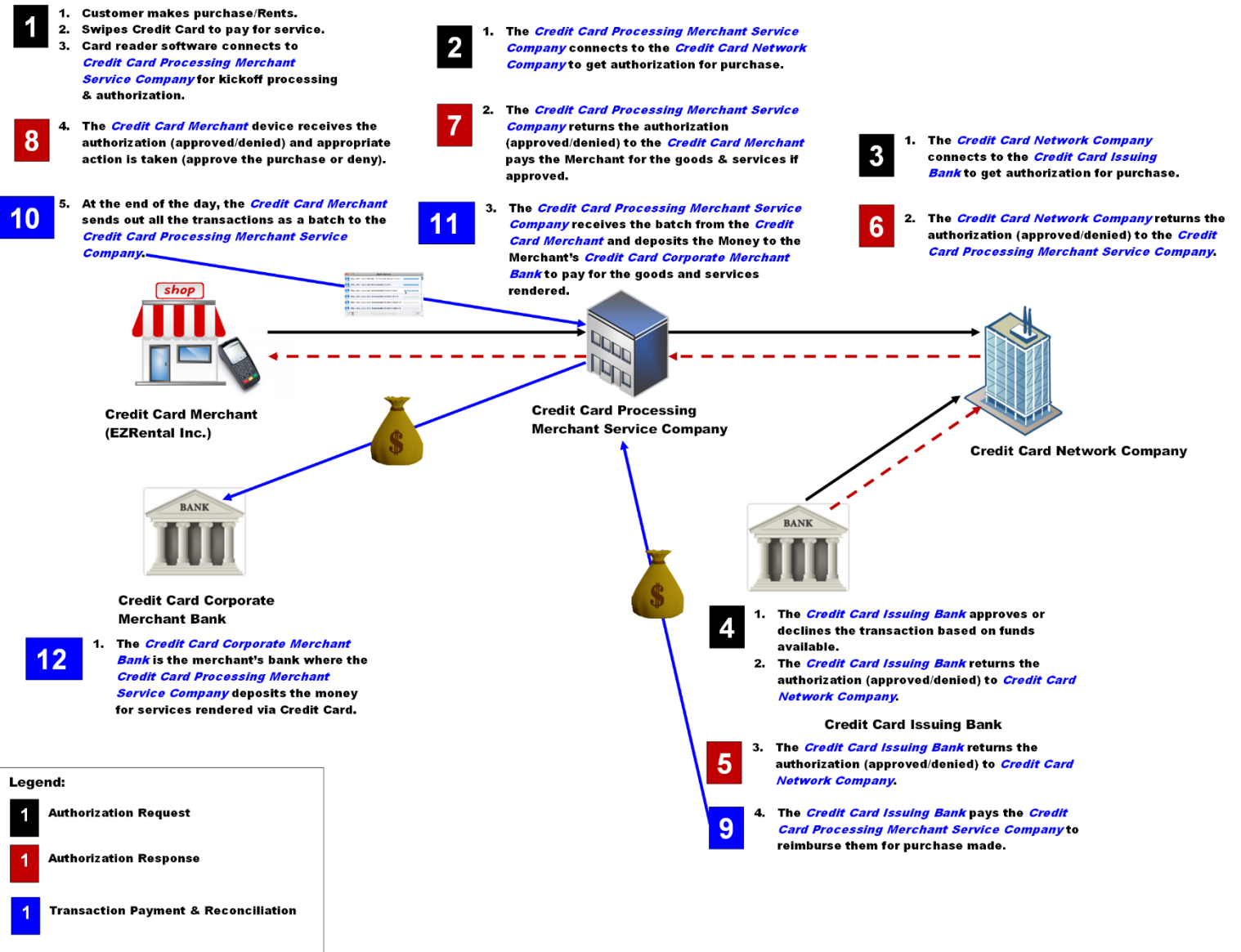
- **Credit Card Corporate Merchant Bank Code**– In our business, we use and store a number code used to identify the *Credit Card Corporate Merchant Bank*. This code has business meaning and appears in reports and discussed by users. Therefore, a *Credit Card Corporate Merchant Bank* can be identified by this code.
- **Credit Card Corporate Merchant Bank Name**– In our business, we also use and store the name of the *Credit Card Corporate Merchant Bank*. This name also has business meaning and appears in reports and discussed by users. Therefore, a *Credit Card Corporate Merchant Bank* can be identified by its name.
- Below is a listing of the values/instances of the *Credit Card Corporate Merchant Bank Code* and *Credit Card Corporate Merchant Bank Name* we use at **EZRental Inc.**:

<i>Credit Card Corporate Merchant Bank Code</i>	<i>Credit Card Corporate Merchant Bank Name</i>
1	Chase
2	Citi
3	Capital One

BUSINESS REQUIREMENTS

OUR CUSTOMERS & CREDIT CARD PROCESSING (CONT.):

Below, is a pictorial representation of the interaction between the credit card processing entities **Merchant (EZRental Inc.)**, **credit card processing merchant service company**, **credit card network company** and **credit card issuing bank**:



BUSINESS REQUIREMENTS

OUR CUSTOMERS (CONT.):

Corporate Customers

Corporate Customers are customers who are renting vehicle during business travel and their company have a contract with **EZRental Inc.** These companies get special corporate rate for their employee's rental services. Therefore, for our **corporate customers only**, we must store the following attributes/properties: unique **company ID** (we have a unique ID number for each company doing business with us), **company name**, **company address** which contains the elements: **address line1**, **address line 2** (which is optional and used for apartment number, suite or any additional address information required), **city**, **state code**, **zip code** (which is the two-character code for a state in the US) & **country**, in addition, **company contact** which is composed of **company representative name**, **contact phone number** & **contact email** (unique as all email addresses). And finally, we need to store the **company discount percentage rate** which is the discounted percentage applied to a corporate customers rental. The company Discount percentage rate is store in the database as a decimal percentage value, for example 20% is stored as 0.20, 30% as 0.30. 50% as 0.50 etc. This discount percentage (0.0x) is applied to the **Vehicle Rental Categories** which determines the price of each category to determine the total discount. Therefore, when a corporate customer rents a vehicle from a vehicle category (such as economic, compact, standard etc.), this discount percentage is applied to each of the categories during the rental/reservation process. Note that every company has a different percentage rating depending on their contract with **EZ-Rentals Inc.** For example, some companies have 20% discount towards their rentals, which would be stored as 0.20 in the database, some have 30% (0.30) etc. Vehicle Rental Categories are discussed in more details later in these requirements.

Retail Customers

Retail Customer Discounts

Retail Customers can (but don't have to) leverage promotional **discounts** or multiple coupons obtain from other businesses, internet, magazine, organizations, etc., to save money on their rentals. Therefore, we need to capture specific data for the promotional discounts used by a retail customer. A **Promotional Discount** is composed of the following attributes: **discount ID**, a unique random number which uniquely identifies a discount, another unique **discount code** or the coupon code itself used to redeem the coupon, which is an alphanumeric code **10-characters** long. This code is generated by our marketing team and published to magazines, newspapers, internet e-commerce sites, etc. Finally, the last attribute is **discount code description** or description of the discount. Examples of currently used **discount ID**, **discount code**, **discount code description** are shown in table below:

Discount ID	Discount Code	Discount Code Description
1234..	AAA9970054	AAA Membership Discount - 25% off base rate plus 10% donated for breast cancer research.
5678..	GOV8756921	Government Employee Discount - 30% off base rate
9101..	STA3415632	State Employee Discount for 25% off base rate
1213..	VET2055179	Veteran Discount 35% off base rate Plus 10% donation to veteran's family fund.
Etc..	Etc..	Etc..

BUSINESS REQUIREMENTS

OUR CUSTOMERS (CONT.):

Retail Customer EZPlus Rewards Program

Retail customers can opt-in to enrolled in the **EZPlus Rewards Program** where they earn points for every rental of a vehicle. These rewards points can be redeemed for future rentals. Note that the **EZPlus Rewards Program** is **optional** for retail customers & points are earned only when they rent vehicles. For the **EZPlus Rewards Program** we need to store unique random number **EZPlus ID**, the unique **Ezplus rewards code** which is the code used in the business when managing the **EZPlus Rewards Program**. This random code is generated and assigned to a **Retail Customer** by the client application. The number starts with the 3-characters EZP and a 10-digit number e.g., **EZP9999999999**, and the final attribute is the **EZPlus rewards earned points**, which is an integer that indicates the number of rewards points earned that accumulated after all the rentals and can be used to save on future rentals. **Examples of currently used EZPlus ID, EZPlus rewards Code and EZPlus earned points that we currently use are:**

<i>EZPlus ID</i>	<i>EZPlus Rewards Code</i>	<i>EZPlus Rewards Earned Points</i>
1234..	EZP9009854637	10000
5678..	EZP1000192461	500
9101..	EZP6493238865	159000
1213..	EZP2005135627	23000
Etc..	Etc..	Etc..

The following business rules apply to the EZPlus Rewards Program:

1. *Only a Retail Customer can leverage the **EZPlus Rewards Program**, no other type of customer such as a Corporate Customer can join the EZPlus program.*
2. *The **EZPlus Rewards Program** is OPTIONAL. A Retail Customer can join the **EZPlus Rewards Program** during registration or any other time after or not join at all.*
3. *A Retail Customer can drop out of the EZPlus Rewards program at any time.*
4. *Every time a Retail Customer that is member of the **EZPlus Rewards Program** rents a vehicle, they earn **1000 EZPlus Rewards Points**. When a Retail Customer member of the **EZPlus Rewards Program** earns **10,000 EZPlus Rewards Points** they earn a FREE RENTAL.*

As an incentive for our retail customers to join the **EZPlus Rewards Program** during registration to become a customer, we offer a **EZPlus sign-up rewards points** of **1000 EZPlus Rewards Earned Points**. Also note that the maximum number of **EZPlus Rewards Earned Points** is capped at **50,000** points. A Retail Customer cannot accumulate more than **50,000** points in the **EZPlus Rewards** program.

BUSINESS REQUIREMENTS (CONT.)

OUR CUSTOMERS (CONT.):

In this business, we have the following business rules for our customers (*Retail* or *Corporate*):

1. *We only have two types of customers retail customer or corporate customers. No other type of customer exists.*
2. *The minimum age to be a customer and rental our vehicles is 21 years old. This rule must be enforced.*
3. *A customer cannot be a retail & corporate customer at the same time. A customer can only rent as a retail customer or as a corporate and these transactions must be separate. We don't want our customers to be able to combine both retail customer discounts, rewards program and corporate rates at the same time.*

OUR VEHICLES:

EZ-Car Rental needs a system to manage their vehicles for renting, maintenance, selling, etc. Vehicles are classified into 4 main types: **CAR**, **SUV**, **MINIVAN**, and **CARGO VAN**. These are the vehicles most rented and available at every rental agency. Nevertheless, there are other categories of vehicles available only certain rental agency locations such as **RECREATIONAL VEHICLES**, **MOTORCYCLES**, **MOBILE HOMES**, etc. No matter what type of vehicle being rented, all vehicle types share the following common characteristics:

- Each vehicle is identified by the random number **vehicle ID**. In addition, each vehicle is also identified by the alpha-numeric **vehicle VIN number**. Note the following business rule on a **vehicle VIN number**:

1. *The vehicle VIN number is used throughout the business to identify a vehicle for searching, reporting etc.*
2. *Therefore, the vehicle VIN number is the unique ID for a vehicle to be identified and managed from a business perspective.*

- Other attributes include the **vehicle name** composed of **make**, **model** & **year**. Additional attributes are **color**, also the **license plate** composed of the following components: **license plate number**, **license plate state**.
- More attributes are **mileage**, **transmission type** of the vehicle. The Transmission Type attribute has business value thus used in reports and in the business processes. The values used for **transmission type** and a **transmission type description** as follows:

Transmission Type	Transmission Type Description
1	Manual Transmission
2	Automatic Transmission
3	Continuously Variable Transmission (e.g., CVT).
4	Semi-automatic Transmission
5	Dual-clutch Transmission
6	Transaxle Transmission

Vehicle Status ID	Vehicle Status Description
1	Available
2	Reserved
3	Rented
4	Not available
5	Maintenance (Not available)
6	Dropped off and located at another agency
7	In Transport to Owning Agency
8	No Longer available for rental

- **seat capacity** attribute, which is the number of seats in the vehicle. Vehicles such as *cars* have a seat capacity of 5 passengers (2 in front and 3 in the back), *SUVs* have 7 or 8 passengers. Cargo Vans have only 2 passenger seat capacity, Minivan have 8 to 9 passengers, special vehicles such as passenger van hold 12 passenger seat capacity, a shuttles bus can hold 16 to 20 passengers, mini-buses 30 to 40 passengers and large busses can hold 70 passengers.
- All vehicles also have a special code and description that we use to track the vehicle status named **vehicle status ID**. This is a unique number that identifies the status of a vehicle, which works in conjunction with **vehicle status description** which describes the status represented by the **Vehicle Status ID**, such as **reserved**, **rented**, **available**, **maintenance**, **not available**, **transferred**, etc. Below is the list of vehicle status IDs we are currently using and their descriptions:

BUSINESS REQUIREMENTS (CONT.)

OUR VEHICLES (CONT.):

In addition to these attributes shared by all vehicles, there are 4 main categories of vehicle which share unique characteristics than the other types of vehicles found in our agencies. These 4 types are as follows:

- A **Car** is a vehicle whose **trunk capacity** (measured in cubic feet volume) is advertised to our customers. Customers can decide which vehicles better fits their needs based on the trunk capacity and number of luggage they are carrying etc. For example, a *luxury Mercedes E class* car has a trunk capacity of 18.5 cubic ft., which has a large trunk capacity.
 - An **SUV** is a vehicle with a **towing capacity** attribute in pounds. Towing capacity is a single number in pound or could also be a decimal number in pounds. For example, some of our SUV have a maximum towing capacity of 3,000 pounds etc. Another attribute of SUV is an attribute classification if the SUV is **All-Wheel-Drive**, which stores a Boolean value of **YES/NO** or **TRUE/FALSE**.
 - A **Minivan** has the option of **having a disability package**, which is also a Boolean value of **YES/NO** or **TRUE/FALSE**.
 - Finally, a **Cargo Van**, has a **cargo capacity** in cubic feet volume. For example, the typical volume of our Vans is 245 cubic feet (cu.ft.). Cargo Vans also have a **maximum payload** attribute that determines how much weight in pound it can hold. Our cargo vans have typically a maximum payload of 3,880 lbs.
- As stated previously, there are other types of vehicles of interest that in some location we may want to store data on other than car, SUV minivans and cargo van.
- Note that the following Business Rules were identified by the business stakeholders on the vehicles:

1. *A reservation/rental can only be for one of these four categories of Vehicles or other vehicle types, not a combination.*
2. *This means, you can only rent either a car, SUV minivans, cargo van or other for a reservation or rental, not a combination such as a car & SUV at the same time. Each reservation is unique to one vehicle.*

Below are additional business rules for our vehicles and agency ownership:

1. *Every vehicle is owned by one agency. The vehicle can be pick-up and dropped-off at any agency, but only one agency is the vehicle's owning agency. An agency can own many vehicles, but a vehicle can only be owned by one agency.*
2. *A vehicle can currently be located at any agency depending on where it was dropped-off after a rental. We need to track the current agency where the vehicle is located, to arrange a transfer or a rental that will ultimately direct the vehicle to the owning agency.*

RESERVATION PROCESS:

A vehicle must be reserved if a customer wants to guarantee the vehicle will be available for rental. There is a distinction between a reservation and a rental. A reservation guarantees a vehicle will be ready for you to be pick-up and rented. A rental means a customer complied with the reservation and rented the vehicle. On the other hand, a customer can walk into an agency and rent without reservation but only vehicles that are available at the time and not reserved.

We have the following business rules for reserving a vehicle reservation:

1. *A reservation is NOT made for a specific vehicle, but to a vehicle rental category. Rental category examples are economy, intermediate, full size, luxury.*
2. *Thus, a customer makes a **reservation** of a **vehicle rental category** at a **rental agency**. Therefore, the reservation process involves a customer a vehicle rental category and the rental agency where the vehicle will be picked up.*

BUSINESS REQUIREMENTS (CONT.)

RESERVATION PROCESS (CONT.):

A **Vehicle Rental Category** contains a list of vehicles depending on the vehicle type: Car (economy, intermediate, full size, luxury), SUV (standard, full size etc.), or Cargo Van etc. Each of these categories have a different price range. Therefore, for a vehicle rental category we need to capture the unique **vehicle rental category ID** that identifies the category of the vehicle being reserved or rented, **category name** and finally **category daily rental rate** for the category. We used a specific code for our vehicle rental category ID, category name & daily rental rate. The table below shows the ID, category names and rate we currently using in our business:

Vehicle Rental Category ID	Vehicle Rental Category Name	Category Daily Rental Rate
1	Car-Economic	\$113.99
2	Car-Compact	\$115.99
3	Car-Intermediate	\$116.67
4	Car-Standard	\$119.99
5	Car-Full Size	\$121.99
6	Car-Premium	\$127.79
7	Car-Luxury	\$139.99
8	SUV-Intermediate	\$127.99
9	SUV-Standard	\$128.99
10	SUV-Standard Elite	\$135.99
11	SUV-Full Size	\$148.99
12	SUV-Premium	\$157.99
13	Minivan-Standard	\$152.99
14	Van-Cargo Van	\$19.95
15	Pick Up-Mid Size	\$69.95
16	Pick Up-Full Size	\$105.99
17	Motorcycle-Touring	\$19.95
18	Motorcycle-Cruiser	\$199.99
19	Motorcycle-Scooter	\$79.95
20	Passenger Van (12 passengers)	\$161.00
21	Passenger Shuttle (16 passengers)	\$180.00
22	Passenger Shuttle (20 passengers)	\$220.00
23	Passenger Mini-Bus (30 passengers)	\$250.00
24	Passenger Mini-Bus (40 passengers)	\$280.00
25	Passenger Large-Bus (80 passengers)	\$300.00

We have the following business rule relate to a vehicle and a vehicle rental category:

1. A vehicle is a member of a vehicle rental category.
2. A vehicle rental category can have one, none or many vehicles belonging to that category at any given time, nevertheless, a vehicle can only belong to one vehicle rental category.

As stated previously, a customer makes a reservation of a vehicle rental category at a rental agency. Therefore, the reservation process requires the **customer, vehicle rental category & rental agency** for a reservation to be made. The following business rules apply to a reservation:

1. A vehicle can be reserved to be picked up at the INDICATED rental agency and dropped off at the SAME rental agency.
2. A vehicle can be reserved to be picked up at the INDICATED rental agency and dropped off at a DIFFERENT rental agency.
3. A reservation is made only for one pick-up rental agency, but a rental agency can have many reservations for pick-ups taking place.
4. A reservation can only be for one drop-off rental agency, but a rental agency can have many reservations drop-offs taking place.
5. For reporting and analytics, we need to capture all changes to the reservation's pick-up and drop-off agency, such as all changes by the customer for pick-up agency & drop-off agency later after the original reservation. This means we need to store all history on all reservation pick-and drop-off agency changes.

When a customer reserves a vehicle rental category for a specific rental agency, we wish to capture the following:

- A unique **reservation ID** which is used by the business to manage and track reservations, the **rental agency ID** where the vehicle will be picked up, and the target **reservation drop-off rental agency**.
- In addition, we need to store the **reservation schedule** composed of **reservation pick up date**, **reservation pick up time**, **reservation drop off date** and **reservation drop off time**, also the **reservation estimated rental cost**. A reservation can have multiple schedule changes during the lifetime of the reservation since customers can make changes to the reservation and we need to track this history of changes for analytical purposes.

BUSINESS REQUIREMENTS (CONT.)

RESERVATION PROCESS (CONT.):

We have the following business rule relate to the *reservation schedule*:

- 1. For reporting and analytics, we need to capture all changes to the reservation schedule, such as all changes by the customer for pick-up date & time and drop-off date & time. This means we need to store all history on all reservation schedule changes.
 - 2. A customer can have MANY reservation schedules based on changes to the reservation, but a schedule can only belong to ONE customer.
- Finally, we need to store the unique *reservation status ID* which is a unique number we use to indicate the status of a reservation and *reservation status description* which describe each of the status such as: **confirmed**, **cancelled**, **completed** etc. Below is an example of the *reservation status ID* and *status description* we currently use in our business.

Reservation Status ID	Reservation Status Description
1	Confirmed
2	Modified & reconfirmed
3	Cancelled
4	Fulfilled & closed
Etc..	Etc..

For a reservation we must adhere to the following business rules:

- 1. A customer can make none, one or many reservations for a vehicle rental category at a rental agency.
- 2. A rental category can be reserved by none, one or many customers at a rental agency.
- 3. A rental agency can get many or no reservations for a vehicle rental category by a customer.
- 4. A reservation can only have one pick-up rental agency location, but a rental agency can have many reservation pick-ups happening.
- 5. Each reservation has a drop-off rental agency (may be different than pick-up rental agency). A reservation can only have one drop-off rental agency location, but a rental agency can have many reservation drop-offs taking place.

THE RENTAL PROCESS:

Once a vehicle has been reserved, the vehicle can be rented (picked up/dropped off) as per the scheduled of the reservation agreement. A rental means a customer complied and fulfilled the reservation and rented the vehicle.

For the rental process, the following business rules apply:

- 1. A customer rents a vehicle Rental Category at a rental agency. This means the rental process requires the **customer, vehicle rental category**, and **& rental agency** for a rental to be complete.
 - 2. A Rental includes a specific Vehicle of the vehicle rental category. A vehicle can be rented many times, but a rental is only for one vehicle only. You cannot rent multiple vehicles in one rental contract.
 - 3. During the rental process we may have any of the following business rules/scenarios:
 - 1) A vehicle can be picked up at the **SAME** rental agency as indicated by the reservation and dropped off at the **SAME** rental agency.
 - 2) Or a vehicle can be picked up at the **SAME** rental agency as indicated by the reservation and dropped off at **ANOTHER** rental agency.
 - 3) Or a vehicle can be picked up at **ANOTHER** rental agency other than what was indicated by the reservation and dropped off at **SAME** rental agency of the reservation.
 - 4) A vehicle can be picked up at **ANOTHER** rental agency other than what was indicated by the reservation and dropped off at **ANOTHER** rental agency of the reservation.
 - 5) For reporting and analytics, we need to capture all changes to the rental pick-up and drop-off agency, such as all changes by the customer for pick-up agency & drop-off agency later after the original reservation. This means we need to store all history on all rental pick-and drop-off agency changes.
- ❖ Note that for scenarios 3 & 4, we cannot guarantee that the vehicle rental category of the reservation will be available at the agency other than what was agreed in the reservation. We will do our best to accommodate the change during these scenarios or find another vehicle that will be closed to the original reservation.

For the rental process, the following business rules also apply:

BUSINESS REQUIREMENTS (CONT.)

THE RENTAL PROCESS (CONT.):

- Other attribute we need to capture the **rental deposit** for a rental. The rental deposit value is calculated based on the **rental period + 25% of the rental period** and for any damage or other charges that were incurred during the rental period. This deposit is refunded to the customer's credit card when the vehicle is returned in the condition in which it was rented.
- Finally another attribute we need to capture is the **rental total cost** or total cost that needs to be paid by the customer. This value is calculated based on selected *fuel option, insurance option, vehicle rental category* price and other factor such as such as duration of the rental etc.

We need to be able to associate a reservation to a rental and vice versa, therefore we maintain the following additional business rules for our rental & reservation:

1. *A reservation is made for a rental and the opposite holds true; a rental is based on a reservation.*
2. *But NOT all rentals are based on a reservation. We allow a customer to walk into a rental agency and rent a vehicle without a reservation.*
3. *When a reservation is made for a rental, then it must be for only one rental, and a rental can be for a reservation but not mandatory since a customer can walk into an agency and rent a vehicle without a reservation.*

OUR EMPLOYEES:

EZ-Car Rental currently has 5,500 employees across the world. We do expect to grow as we move into new markets such as Asia, Africa, and the Mediterranean. But our business does not require a large workforce, therefore, we don't expect to grow more than 12,000 in the next 10+ years. Our employees consist of *customer service agents* in the Rental Agencies & online support who interact with our customer to reserve and rent vehicles. In addition, *back-office inventory personnel, auto specialists* who work in our services centers servicing our vehicles, *drivers* to transport our vehicles from one agency to another and *maintenance personnel* who maintain our agencies and finally our *business team* that handles the day-to-day business activities in our agencies and other roles. For now, we are only interested in storing the following data for all these types of employees:

- An **Employee ID** which uniquely identifies the employee, **employee name** which is composed of: *first name, last name*, also **employee address** which includes the components: *address line1, address line 2, city, state code, zip code & country*. Also, **employee phone, employee job title and employee email**. In addition, we need to capture the employee **social security number**. Below are some business rules and usage for the **EmployeeID** and the **social security number**.

The following business rules related to employees must be followed:

1. *The employee social security number needs to be protected and secured as per federal regulations. All security measures such as encryption, etc., need to be taken to protect the social security number; therefore, the full social security number **cannot** be seen by employees, reports, and other business processes.*
2. *In special cases where the social security number needs to be displayed, only the last 4 digits will be shown using the following format *****_**_1234**. Nevertheless, the goal is **NOT** to display the social security number as much as possible, and it should only be used internally within the application for processing but not displaying.*
3. *The EmployeeID number is what is used throughout the business to identify an employee for searching, reporting. business processing, etc., therefore, the EmployeeID is the unique ID for an employee to be identified and managed from a business perspective.*
4. *The minimum age to be an employee of our company 18 years old. This rule must be enforced.*

SECURITY & APPLICATION ACCESS:

To access our systems proper security and authentication is required. Only authorized users can have access our agencies Point-Of-Sales & Back-End Management systems. In addition to our **EZRental.com** portal by our customers. Therefore, due to security and regulatory compliance purpose, we want to separate the employee access data from the customer access data by using two separate user accounts:

- Employee user accounts
- Customer user accounts

BUSINESS REQUIREMENTS (CONT.)

Security Access for our Customers who register for our EZ-CarRental.com web site (Customer User Accounts):

Customer who accesses our online portal to reserve and rent our vehicles also need a username and password to access our system, therefore each customer owns a customer user account.

A customer user account should store the user **customer user account ID** a unique alpha-numeric string identifier that identifies the customer user account, **customer username** another unique alpha-numeric value that identifies each customer, the **customer password** that is an alpha-numeric known only to the customer, and finally, the customer **email** to map the customer user-account to a customer. Note the following business rule:

- 1. A customer can own one customer user account only, and a customer user account can only be owned by one customer.
- 2. For a period of time, we will need to register customers into our **EZRental.com** business, nevertheless the web portal may NOT be implemented or completed when new customers are registering at this time, therefore, for period of time, creating a customer user account when registering a new customer is optional until the Web Portal Application is created. But is important in the future, that we force the creation of customer user accounts when a new customer is registered once the Web Portal Application is ready. It is the responsibility of the database architect(s) and full-stack developers to update this feature when the appropriate time comes.

VEHICLE TRANSPORTATION:

We need to know where our vehicles are located at all times, such as at the Rental Agency that owns the vehicle, another Rental Agency that does not own the vehicle, being transported from one Rental Agency to another as a result of a vehicle transfer after a rental to the owning rental agency, being transported as a new delivery to a Rental Agency from our distribution center, being transported for maintenance, or currently being rented by a customer. Vehicles need to be tracked or location status known. At this time, we are only interested in tracking when a vehicle is transported from one Rental Agency to another Rental Agency under the following scenarios:

- Vehicle can be located at a Rental Agency that does not own the vehicle after a rental dropping off at a different location than the picked up owning Rental Agency, thus vehicle eventually needs to be transported and delivered to the owning agency.
- Another non-owning Rental Agency requests support from other Rental Agency(s) for loans of vehicle(s) to borrow due to an unexpected busy period and requesting agency is short on inventory. After the first agency is done with the loaner vehicles, these vehicles need to be returned to the borrowed owning Rental Agency(s).
- In our current process & systems we currently use the following reason IDs and reason descriptions:

Transport Reason ID	Transport Reason Description
1	Rental Drop off at different location
2	Vehicle Loaned to another Agency
3	Pick up from Distribution Center
4	Drop off to Distribution Center
5	Vehicle sent for maintenance
7	Unknown

Transport Status ID	Transport Status Description
1	Transport completed
2	On route to pick up location.
3	On route from pick up location
4	At pickup location. In progress (Loading etc.)
5	Pickup location delay
7	Unknown

Note that transportation to and from Rental Agency is executed by an employee who is part of a transportation team or drivers. Therefore, when an employee executes a transport request of a vehicle to and from Rental Agencies, we need to capture the following information:

BUSINESS REQUIREMENTS (CONT.)

CONCLUSION:

The business data listed in this business requirements document is what we need to capture for our business to operate. As our business evolve, additional data will be required in the future. We will address these new requirements in future versions of the application. For example, invoice processing & employee management at our rental agencies are features on our roadmap. Therefore, our expectations are that the design is modular and scalable for future growth.

APPLICATION DEVELOPMENT TECHNICAL REQUIREMENTS

The technical architecture was designed to ensure database security, performance, and scalability. Key requirements included:

- Centralized database instance hosted on Oracle Database 23c Free Edition
- Normalized schema design to Third Normal Form (3NF)
- Primary and foreign key enforcement for referential integrity
- Support for multi-user access and transactional consistency
- Secure customer data handling with appropriate constraints
- Physical schema design aligned to Oracle data types and engine capabilities

This foundation ensured that customer and payment information could be reliably maintained and accessed as the platform scales.

APPLICATION PHYSICAL & SOFTWARE TECHNICAL ARCHITECTURES

The system architecture was implemented on a local workstation environment configured for development and testing. The solution design incorporated:

- Oracle Database 23c Free Edition
- SQL Developer for data modeling, script execution, and schema visualization
- Windows-based development workstation
- Standard three-tier application architecture concept database tier ready for future middleware and presentation tiers

This environment provided the needed capability to build, deploy, and validate database structures within the defined sprint timeline.

APPLICATION DEVELOPMENT FEATURES & FUNCTIONALITIES (AGILE BACKLOG)

Core Sprint Backlog Features:

Feature	Description
Customer Creation	Support onboarding & secure profile creation
Customer Identity Validation	Store ID credentials & structured identity attributes
Payment Profile Storage	Maintain multiple payment methods per customer
Fraud Prevention Data Rules	Enforce customer verification before activation
Data Normalization	Ensure structured data format to 3NF
ER/EER Model Creation	Build conceptual architecture of system entities
Logical Model Development	Generate normalized schema & relationship definitions
Project Documentation	Deliver enterprise-style business and technical documents

Future backlog items (not in Sprint 1):

- Rental transaction management
- Vehicle fleet and availability
- Branch operations
- Web reservation system integration

DATABASE MANAGEMENT SYSTEM DEVELOPMENT ENVIRONMENT & PHYSICAL ARCHITECTURE

The database system was implemented and validated using Oracle Database 23c Free Edition installed locally on a development workstation. SQL Developer was used as the primary interface for schema modeling, execution, and inspection.

The architecture was designed to support a scalable relational data environment, allowing future deployment on cloud or enterprise server environments. The physical design followed standard relational practices with defined PK/FK constraints, data types, and normalization standards.

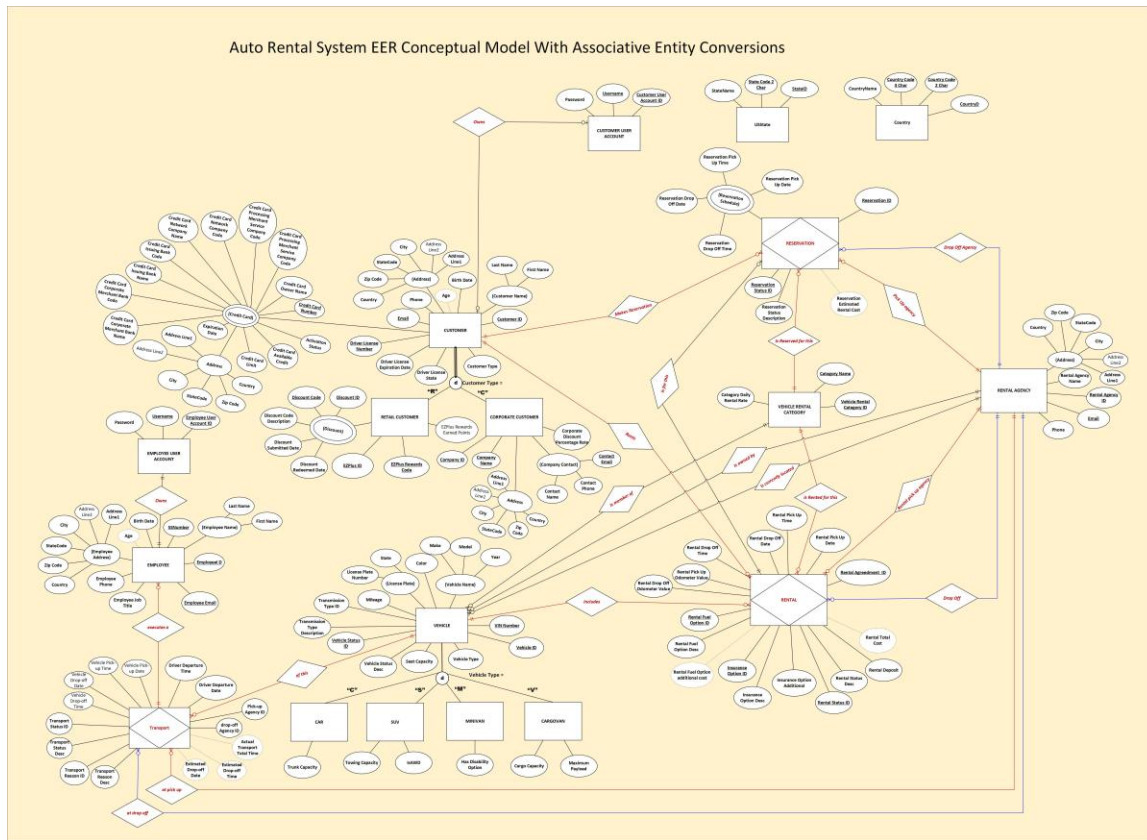
PROJECT ROLES & RESPONSIBILITIES

The EZRental Auto Rental POS database project was developed by a cross functional team following a consulting model, where responsibilities were clearly defined to ensure accountability, quality, and timely delivery. The development activities included requirement analysis, data modeling, database design, documentation, and implementation preparation. The table below outlines the roles and responsibilities associated with this project phase.

Project Roles Table

Role	Responsibility
Client EZRental, Inc.	Defined enterprise rental business needs, provided domain expectations, and approved project objectives.
Project Sponsor EZRental Executive Team	Ensured project alignment with corporate vision, approved scope, and supported enterprise adoption.
Project Manager Prof. Abel Rodriguez	Provided direction, evaluated deliverables, managed academic compliance, and ensured methodology adherence.
Lead Database Consultant Alisher Ismoilov	Analyzed requirements, designed conceptual & logical data models, documented system specifications, and prepared initial implementation assets.
Business Analyst	Interpreted business processes, facilitated requirement translation to data structures, and guided conceptual model definition.
Systems Architect	Referenced for future phases responsible for defining long term technical architecture and integration approach.

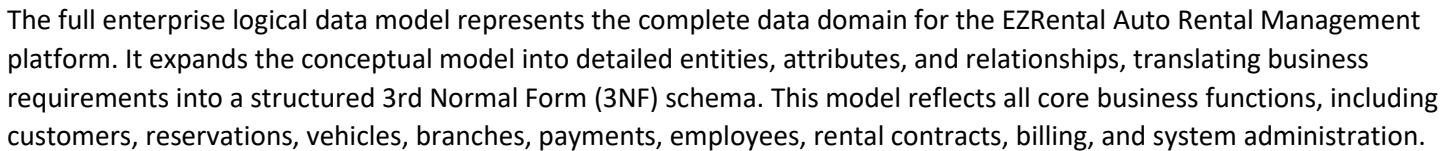
ER/EER CONCEPTUAL MODEL DIAGRAM



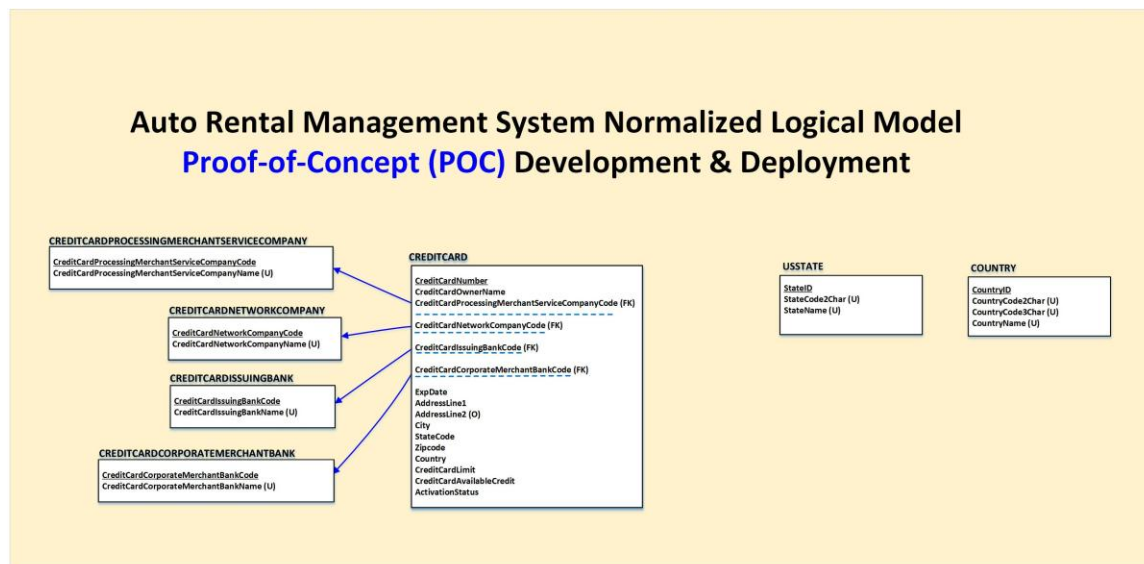
The ER/EER conceptual model represents the high level data architecture for the EZRental Auto Rental POS Management System. This model was developed during the conceptual design phase to capture business-level entities and relationships before normalization and physical implementation activities began. The diagram reflects the core domains of the rental business, including customers, payment profiles, rental operations, vehicles, branch locations, and associated business entities.

The conceptual model abstracts data at the enterprise level, showing how key entities interact to support EZRental’s operational needs across rental branches, corporate teams, and online customer platforms. This stage served as a blueprint for transforming business requirements into a structured database design, ensuring alignment with enterprise rental workflows, fraud prevention rules, multi location operations, and future scalability needs prior to logical modeling and physical schema development.

NORMALIZED LOGICAL MODEL DIAGRAM



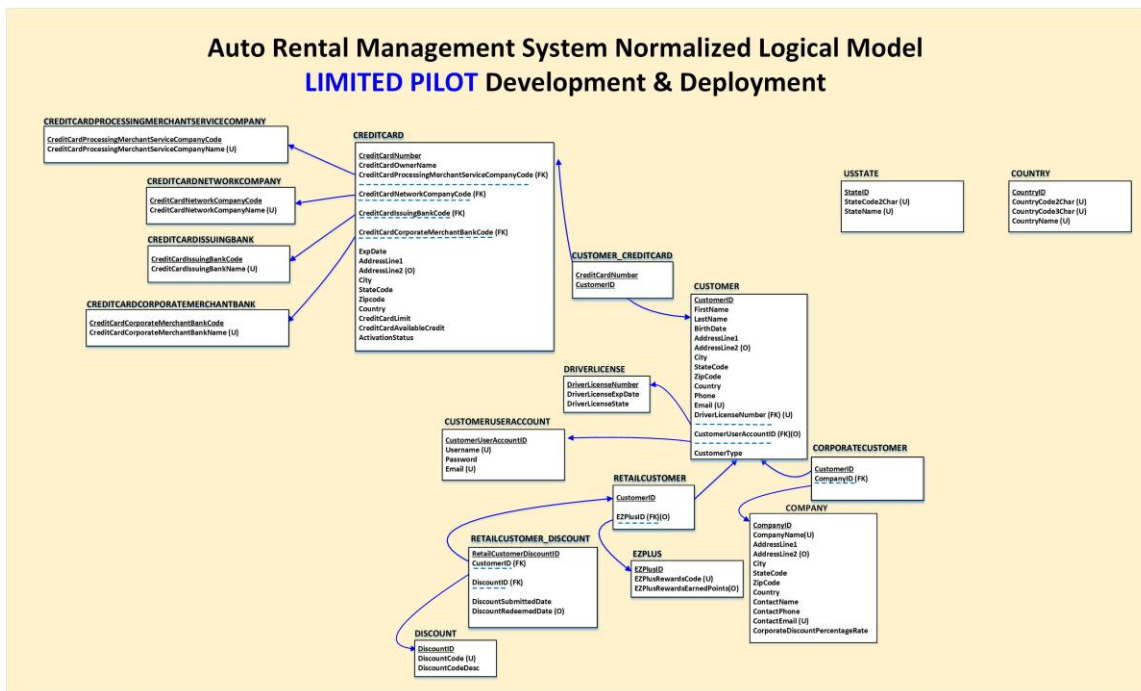
This enterprise level model illustrates the full scope of EZRental’s future operational database once all platform modules are implemented. Although only a subset of entities will be built in the pilot stage, the complete logical model ensures long term scalability, alignment with enterprise architecture standards, and supports future expansion into global markets, online services, and analytics capabilities.



ROJECT 1 – EZRental Auto Rental POS Management System Database Design and Implementation

The Proof-of-Concept (POC) logical model focuses on the core foundational tables required to validate fundamental database structure and design principles. This simplified schema contains only the essential entities needed to demonstrate support for customer creation, identity management, and basic payment capability.

The objective of this model was to validate entity definitions, primary and foreign key structure, normalization rules, and initial system workflows. The POC stage ensured that the core data elements were properly defined and that a scalable structure existed prior to expanding into the full pilot database. This model served as an internal checkpoint to confirm design accuracy before building the larger limited pilot schema.



The limited pilot logical data model represents the finalized subset of the enterprise schema that will be implemented in the initial build phase. This 17-table schema supports customer profile creation, multi-card payment profile management, regional and branch association, risk/fraud controls, and operational readiness for integration with reservation and rental modules in future project phases.

This pilot model forms the foundation of the working database, ensuring that EZRental can securely store, validate, and manage key customer and payment data across physical branch locations and online systems. It was designed to uphold data integrity, performance, and security standards while supporting future system expansion and real-world deployment.

PHYSICAL MODEL DATA DICTIONARY

#1:

CREDITCARDPROCESSINGMERCHANTSERVICECOMPANY							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length/Size /Format	Constraints	Description/ purpose
1.	CreditCardProcessingMerchant ServiceCompanyCode	Number	NUMBER(2)	Yes	20	PRIMARY KEY CHECK(CreditCard Processing Merchant Service CompanyCode between 1 and 20)	Unique Identifier or Primary Key for this table. This Primary Key HAS Business Meaning.
2.	CreditCardProcessingMerchant ServiceCompanyName (U)	String	VARCHAR2(50)	Yes	50	UNIQUE NOT NULL	Stores the unique name of the Credit Card Processing Merchant Service Company. This column value is required and cannot be empty.

#2:

CREDITCARDNETWORKCOMPANY							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length/Size /Format	Constraints	Description/ purpose
1.	CreditCardNetworkCompanyCode	Number	NUMBER(2)	Yes	20	PRIMARY KEY CHECK(CreditCard Network CompanyCode between 1 and 20)	Unique Identifier or Primary Key for this table. This Primary Key HAS Business Meaning.

ROJECT 1 – EZRental Auto Rental POS Management System Database Design and Implementation

2.	CreditCardNetworkCompanyName (U)	String	VARCHAR2(50)	Yes	50	UNIQUE NOT NULL	Stores the <i>unique</i> name of the Credit Card Network Company. This column value is required and cannot be empty.
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#3:

CREDITCARDISSUINGBANK							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length /Size/Format	Constraints	Description/ purpose
1.	CreditCardIssuingBankCode	Number	NUMBER(2)	Yes	1 to 20	PRIMARY KEY CHECK(CreditCard Issuing BankCode between 1 and 20)	<i>Unique Identifier</i> or Primary Key for this table. This Primary Key HAS Business Meaning .
2.	CreditCardIssuingBankName (U)	String	VARCHAR2(50)	Yes	50	UNIQUE NOT NULL	Stores the <i>unique</i> name of the Credit Card Processing Merchant Service Company. This column value is required and cannot be empty.

#4:

CREDITCARDCORPORATEMERCHANTBANK							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length/Size /Format	Constraints	Description/ purpose
1.	CreditCardCorporateMerchantBankCode	Number	NUMBER(2)	Yes	1 to 10	PRIMARY KEY CHECK(CreditCard Corporate Merchant BankCode between 1 and 10)	<i>Unique Identifier</i> or Primary Key for this table. This Primary Key HAS Business Meaning .

ROJECT 1 – EZRental Auto Rental POS Management System Database Design and Implementation

2.	CreditCardCorporateMerchantBankName (U)	String	VARCHAR2(50)	Yes	50	UNIQUE NOT NULL	Stores the unique name of the Credit Card Corporate Merchant Bank. This column value is required and cannot be empty.
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#5:

CREDITCARD							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required ?	Length/ Size /Format	Constraints	Description/ purpose
1.	<u>CreditCardNumber</u>	String	VARCHAR2(16)	Yes	16	PRIMARY KEY	Unique Identifier or Primary Key for the CreditCard table. This Primary Key HAS Business Meaning .
2.	CreditCardOwnerName	String	VARCHAR2(100)	Yes	100	NOT NULL	Credit Card Owner first, last and middle initial if included in name
3.	CreditCardProcessingMerchantServiceCompanyCode (FK)	Number	NUMBER(2)	Yes	20	CHECK (CreditCardProcessingMerchantServiceCompanyCode between 1 and 20) NOT NULL	Column stores the Credit Card Processing Merchant Service Company unique code. Foreign Key to <i>CreditCardProcessingMerchantServiceCompany</i> Table.
4.	CreditCardNetworkCompanyCode (FK)	Number	NUMBER(2)	Yes	1 to 20	CHECK (CreditCardNetworkCompanyCode between 1 and 20) NOT NULL	Stores the Credit Card Network Company unique code. Foreign Key to <i>CreditCardNetworkCompany</i> Table. This column value is required and cannot be empty.
5.	CreditCardIssuingBankCode (FK)	Number	NUMBER(2)	Yes	1 to 20	CHECK (CreditCardIssuingBankCode between 1 and 20) NOT NULL	Stores the Credit Card Issuing Bank unique code. Foreign Key to <i>CreditCardIssuingBank</i> Table. This column value is required and cannot be empty.

ROJECT 1 – EZRental Auto Rental POS Management System Database Design and Implementation

6.	CreditCardCorporateMerchantBankCode (FK)	Number	NUMBER(2)	Yes	1 to 10	CHECK(CreditCard Corporate Merchant BankCode between 1 and 10) NOT NULL	Stores the Credit Card Corporate Merchant Bank unique code. Foreign Key to <i>CreditCardCorporateMerchantBank</i> Table. This column value is required and cannot be empty.
7.	ExpDate	Date	DATA	Yes	MM-DD-YYYY	NOT NULL	The date the credit card expires.
8.	AdressLine1	String	VARCHAR2(50)	Yes	50	NOT NULL	Stores house/building number & street (part 1 of address).
9.	AddressLine2 (O)	String	VARCHAR2(50)	No	50	NULL	Optional value that stores remaining part of address such as apartment number, or other address information.
10.	City	String	VARCHAR2(50)	Yes	50	NOT NULL	Stores the city name
11.	StateCode	Characters	CHAR(2)	Yes	2	NOT NULL	Stores the U.S. State 2-character code. E.g., NY, NJ, CT, etc.
12.	Zipcode	String	VARCHAR2(10)	Yes	10	NOT NULL	Stores US Zip Code/Postal Code. Support format: xxxxx-xxxx.
13.	Country	String	VARCHAR2(100)	Yes	100	NOT NULL	Stores the name of the country.
14.	CreditCardLimit	Number	NUMBER(8,2)	Yes	X = 8 Y = 2	NOT NULL	The maximum amount of dollars that can be charged to the credit card. Assumes that the credit card has the full limit available. The format is NUMBER(X,Y) , where X = The total number of digits and Y = total number of digits to the right of the decimal point. We assume the maximum number that can be stored is 999999.99 for a maximum limit amount \$999,999.99, since we don't expect a customer to have a credit limit of \$1 Million dollars, we cap it at \$999,999.99.
15.	CreditCardAvailableCredit	Number	NUMBER(8,2)	Yes	X = 8 Y = 2	NOT NULL	Stores the current remaining credit available for charging.
16.	ActivationStatus	Boolean	NUMBER(1)	Yes	1	CHECK (ActivationSatus IN (1,0)) NOT NULL	Stores a Boolean value indicating True if credit card is active or False otherwise. Since ORACLE SERVER DBMS does not have a Boolean Data Type, we will use the number 1 to represent True and number 0 to represent False . A CHECK Constraint is used to validate if the number value entered is only a 1 or 0 , any other number value will be rejected by DBMS .

ROJECT 1 – EZRental Auto Rental POS Management System Database Design and Implementation

#6:

CUSTOMER							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length /Size /Format	Constraints	Description/ purpose
1.	CustomerID	Number	NUMBER(10) GENERATED ALWAYS AS IDENTITY	Y	Default size of data type	PRIMARY KEY	Unique Identifier or Primary Key for this table. This Primary Key HAS NO Business Meaning thus an IDENTITY .
2.	FirstName	String	VARCHAR2(50)	Yes	50	NOT NULL	Stores customer's first name.
3.	LastName	String	VARCHAR2(50)	Yes	50	NOT NULL	Stores customer's last name.
4.	BirthDate	Date	DATE	Yes	MM-DD-YYYY	NOT NULL	Stores customer's date of birth.
5.	AddressLine1	String	VARCHAR2(50)	Yes	50	NOT NULL	Stores house/building number & street (part 1 of address).
6.	AddressLine2 (O)	String	VARCHAR2(50)	No	50	NULL	Optional value that stores remaining part of address such as apartment number, or other address information.
7.	City	String	VARCHAR2(50)	Yes	50	NOT NULL	Stores the city name
8.	StateCode	Characters	CHAR(2)	Yes	2	NOT NULL	Stores the U.S. State 2-character code. E.g., NY, NJ, CT, etc.
9.	Zipcode	String	VARCHAR2(10)	Yes	10	NOT NULL	Stores US Zip Code/Postal Code. Support format: xxxxx-xxxx.
10.	Country	String	VARCHAR2(100)	Yes	100	NOT NULL	Stores the name of the country.
11.	Phone	String	VARCHAR2(20)	Yes	20	NOT NULL	Stores customer's phone number.
12.	Email (U)	String	VARCHAR2(100)	Yes	100	UNIQUE, NOT NULL	Stores unique customer email.
13.	DriverLicenseNumber (FK) (U)	String	VARCHAR2(25)	Yes	25	NOT NULL	Stores the customer's Driver License Number. Foreign Key to <i>DriverLicense</i> Table.
14.	CustomerUserAccountID (FK) (O)	Binary	RAW(16)	No	16	NULL	Stores the optional customer's User Account ID which is a GUID . Foreign Key to <i>CustomerUserAccount</i> Table.
15.	CustomerType	Character	CHAR(1)	Yes	1	CHECK(CustomerType IN ('R','C')) NOT NULL	The goal of this column is to stores either the character 'R' representing a <i>Retail Customer</i> or the character 'C' representing a <i>Corporate Customer</i> . A CHECK Constraint is used to validate if character value entered is only an 'R' or 'C', any other value will be rejected by DBMS.

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#7:

CUSTOMER_CREDITCARD							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length /Size /Format	Constraints	Description/ purpose
1.	<u>CreditCardNumber</u>	String	VARCHAR2(16)	Yes	16	PRIMARY KEY	Unique Identifier part 1 of the COMPOSITE Primary Key for this table. Note that together with the <i>CustomerID</i> column, both columns make up the COMPOSITE Primary Key to this table. This column is a Foreign Key to <i>CreditCard</i> Table. Note that the <i>CreditCard</i> parent Table CreditCardNumber Primary Key HAS Business Meaning .
2.	<u>CustomerID</u>	Number	NUMBER(10)	Yes	10	NOT NULL	Unique Identifier part 2 of the COMPOSITE Primary Key for this table. Note that together with the <i>CustomerID</i> column, both columns make up the COMPOSITE Primary Key to this table. This column is a Foreign Key to <i>Customer</i> Table. Note that the <i>Customer</i> parent Table CustomerID Primary Key HAS NO Business Meaning .

#8:

RETAILCUSTOMER							
Column Num	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length/Size /Format	Constraints	Description/ purpose
1.	<u>CustomerID</u>	Number	NUMBER(10)	Yes	Default Size of Data Type	NOT NULL	Unique Identifier or Primary Key for this table. This column is a Foreign Key to <i>Customer</i> Parent Table and a Primary Key to this <i>RetailCustomer</i> Child Table as part of the Supertype/Subtype Structure between <i>Customer</i> Parent Table and <i>RetailCustomer</i> Child Table. Note that the <i>Customer</i> parent Table CustomerID Primary Key HAS NO Business Meaning .
2.	EZPlusID (FK) (O)	Number	NUMBER(10)	No	10	NULL	Stores the optional EZPlusID Foreign Key to <i>EZPlus</i> Table Primary Key EZPlusID . Note that the <i>EZPlus</i> parent Table EZPlusID Primary Key HAS NO Business Meaning .

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#9:

DISCOUNT							
Column Num	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length /Size /Format	Constraints	Description/ purpose
1.	<u>DiscountID</u>	Number	NUMBER GENERATED ALWAYS AS IDENTITY	Y	Default size of data type	PRIMARY KEY	Unique Identifier or Primary Key for this table. This Primary Key HAS NO Business Meaning thus an IDENTITY .
2.	DiscountCode (U)	String	CHAR(10)	Yes	10	UNIQUE NOT NULL	Stores the unique Discount Code value which is a 10 Character string e.g., AAA9970054 as shown in the business requirements.
3.	DiscountCodeDesc	String	VARCHAR2(150)	Yes	150	NOT NULL	Stores the description of the Discount Code listed in business requirements.

#10:

RETAILCUSTOMER_DISCOUNT							
Column Num	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length /Size /Format	Constraints	Description/ purpose
1.	<u>RetailCustomerDiscountID</u>	Number	NUMBER GENERATED ALWAYS AS IDENTITY	Y	Default size of data type	PRIMARY KEY	Unique Identifier or Primary Key for this table. This Primary Key HAS NO Business Meaning thus an IDENTITY .
2.	<u>CustomerID</u> (FK)	Number	NUMBER(10)	Yes	10	NOT NULL	Stores the Customer ID Foreign Key to Customer Table indicating the customer who is registering the Discount.
3.	<u>DiscountID</u> (FK)	Number	NUMBER	Yes	10	NOT NULL	Stores the Discount ID Foreign Key to Discount Table indicating the Discount being registered into the system.
4.	DiscountSubmittedDate	Date	DATE	Yes	Default size of data type	NOT NULL	This column stores the Date the Discount was registered into the system. As per business requirements, in this business during registration a customer can provide a Discount Code during registration to leverage in the future for a rental.

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5.	DiscountRedeemedDate (O)	Date	DATE	No	Default size of data type	NULL	This column stores the Date the Discount was used to rent and get discount on a vehicle rental.
----	--------------------------	------	------	----	---------------------------	------	---

#11:

EZPLUS							
Column Num	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length /Size /Format	Constraints	Description/ purpose
1.	<u>EZPlusID</u>	Number	NUMBER GENERATED ALWAYS AS IDENTITY	Yes	Default size of data type	PRIMARY KEY	Unique Identifier or Primary Key for this table. This Primary Key HAS NO Business Meaning thus an IDENTITY .
2.	EZPlusRewardsCode (U)	String	CHAR(13)	Yes	13	UNIQUE, NOT NULL	Stores the unique EZPlus Rewards Code value which is a 13 Character string e.g., EZP9009854637 as shown in the business requirements.
3.	EZPlusRewardsEarnedPoints (O)	Number	NUMBER	Yes	50000	CHECK(EZPlusRewardsEarnedPoints between 1 and 50000) NULL	Stores the EZPlus Rewards Earned Points accumulated by the customer and available to leverage to get a free rental after 10,000 points as per business requirements.

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#12:

COPRORATECUSTOMER							
Column Num	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length /Size /Format	Constraints	Description/ purpose
1.	<u>CustomerID</u>	Number	NUMBER(10)	Yes	Default Size of Data Type	NOT NULL	Unique Identifier or Primary Key for this table. This column is a Foreign Key to <i>Customer</i> Parent Table and a Primary Key to this <i>CorporateCustomer</i> Child Table as part of the Supertype/Subtype Structure between <i>Customer</i> Parent Table and <i>CorporateCustomer</i> Child Table. Note that the <i>Customer</i> parent Table <i>CustomerID</i> Primary Key HAS NO Business Meaning .
2.	CompanyID (FK)	Number	NUMBER(10)	Yes	10	CHECK(CompanyID between 1 and 20000) NOT NULL	Stores the <i>CustomerID</i> Foreign Key to <i>Company</i> Table Primary Key <i>CompanyID</i> . Note that the <i>Company</i> parent Table <i>CompanyID</i> Primary Key HAS Business Meaning .

#13:

COMPANY							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length /Size /Format	Constraints	Description/ purpose
1.	<u>CompanyID</u>	Number	NUMBER(5)	Yes	5	PRIMARY KEY CHECK(CompanyID between 1 and 20000)	Unique Identifier or Primary Key for this table. This Primary Key HAS Business Meaning .

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2.	CompanyName (U)	String	VARCHAR2(50)	Yes	50	UNIQUE NOT NULL	Stores the <i>unique</i> company name.
3.	AddressLine1	String	VARCHAR2(50)	Yes	50	NOT NULL	Stores house/building number & street (part 1 of address).
4.	AddressLine2 (O)	String	VARCHAR2(50)	No	50	NULL	Optional value that stores remaining part of address such as apartment number, or other address information.
5.	City	String	VARCHAR2(50)	Yes	50	NOT NULL	Stores the city name
6.	StateCode	Characters	CHAR(2)	Yes	2	NOT NULL	Stores the U.S. State 2-character code. E.g., NY, NJ, CT, etc.
7.	Zipcode	String	VARCHAR2(10)	Yes	10	NOT NULL	Stores US Zip Code/Postal Code. Support format: xxxxx-xxxx.
8.	Country	String	VARCHAR2(100)	Yes	100	NOT NULL	Stores the name of the country.
9.	CompanyRepName	String	VARCHAR2(50)	Yes	50	NOT NULL	Stores first, last & middle name if available for the company's representative.
10.	ContactPhone	String	VARCHAR2(20)	Yes	20	NOT NULL	Stores company representative phone number.
11.	ContactEmail (U)	String	VARCHAR2(100)	Yes	100	UNIQUE NOT NULL	Stores <i>unique</i> email for the company's representative.
12.	CorporateDiscountPercentageRate	Number	NUMBER(3,2)	Yes	X = 3 Y = 2	NOT NULL	Stores the percentage discount rate a corporate customer for this company would receive when renting. This decimal number NUMBER(X,Y) should have a decimal format that stores a decimal value 0.00, for example 0.20 is stored in this column if the discount percentage rate for a corporate customer is 20%.

#14:

DRIVERLICENSE							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length/Size /Format	Constraints	Description/ purpose
1.	<u>DriverLicenseNumber</u>	String	VARCHAR2(25)	YES	Default size of data type	Primary Key	<i>Unique Identifier</i> or Primary Key for this table. This Primary Key HAS Business Meaning .
2.	DriverLicenseExpDate	Date	DATE	Yes	MM-DD-YYYY	NOT NULL	Date driver license expires.
3.	DriverLicenseState	Characters	CHAR(2)	Yes	2	NOT NULL	Stores the U.S. State 2-character code. E.g., NY, NJ, CT, etc.

#15:

CUSTOMERUSERACCOUNT							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length/Size /Format	Constraints	Description/ purpose
1.	<u>CustomerUserAccountID</u>	Binary	RAW(16) DEFAULT SYS_GUID()	Yes	Default size of data type	PRIMARY KEY	<i>Unique Identifier</i> or Primary Key for this table. This Primary Key HAS NO Business Meaning and is a GUID generated by the DBMS .
2.	Username (U)	String	VARCHAR2(50)	Yes	50	UNIQUE NOT NULL	The username credentials for a customer user account.
3.	Password	String	VARCHAR2(75)	Yes	75	NOT NULL	The password credentials for a customer user account.
4.	Email (U)	String	VARCHAR2(100)	Yes	100	UNIQUE NOT NULL	Stores <i>unique</i> email tied to this customer user account.

#16:

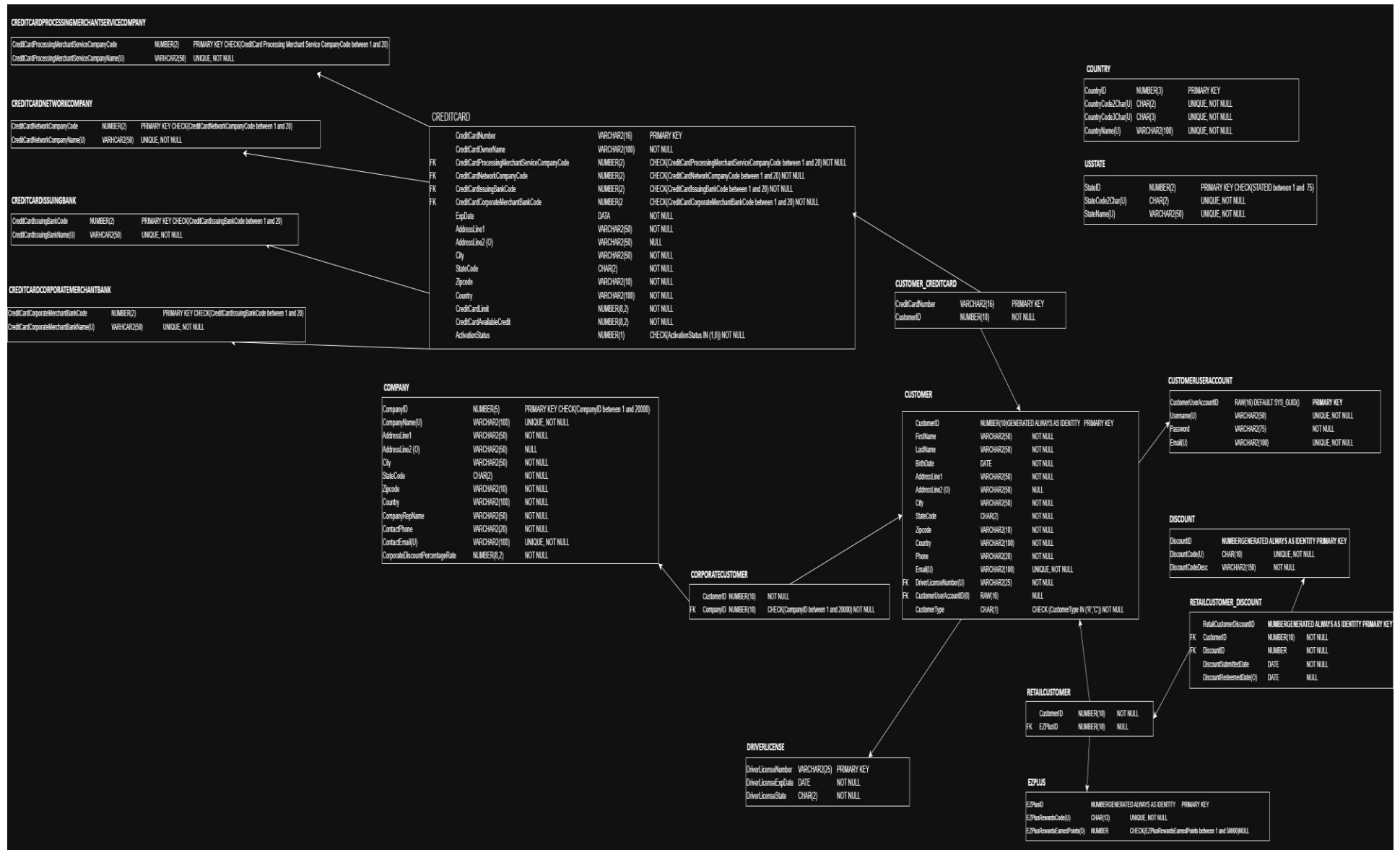
USSTATE							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length/Size /Format	Constraints	Description/ purpose
1.	<u>StateID</u>	Number	NUMBER(2)	Yes	2	PRIMARY KEY CHECK(STATEID between 1 and 75)	<i>Unique Identifier</i> or Primary Key for this table. This Primary Key HAS Business Meaning .
2.	StateCode2Char (U)	Characters	CHAR(2)	Yes	2	UNIQUE NOT NULL	Stores the U.S. State 2-character code. E.g., NY, NJ, CT etc.
3.	StateName (U)	String	VARCHAR2(50)	Yes	50	UNIQUE NOT NULL	Stores the unique full name of the U.S. State.

#17:

COUNTRY							
Column Num.	Attribute/Column Name	Generic Data Type Name	ORACLE SERVER Data Type Name	Is it Required?	Length/Size /Format	Constraints	Description/ purpose
1.	<u>CountryID</u>	Number	NUMBER(3)	Yes	Default size of data type	PRIMARY KEY	<i>Unique Identifier</i> or Primary Key for this table. This Primary Key HAS Business Meaning .
2.	CountryCode2Char (U)	Characters	CHAR(2)	Yes	2	UNIQUE NOT NULL	Stores the country 2-character code. E.g., US, GB, etc.
3.	CountryCode3Char (U)	Characters	CHAR(3)	Yes	3	UNIQUE, NOT NULL	Stores the country 3-character code. E.g., USA, GBR, etc.
4.	CountryName (U)	String	VARCHAR2(100)	Yes	100	UNIQUE, NOT NULL	Stores the unique full name of the country.

ROJECT 1 – EZRental Auto Rental POS Management System Database Design and Implementation

PHYSICAL MODEL SCHEMA DESIGN DIAGRAM



DEVELOPMENT & IMPLEMENTATION

In the Development & Implementation Phase was completed by using Oracle Server DDL (CREATE TABLE) statements to build the physical EZRental database. All tables, primary keys, foreign keys, and constraints from the Physical Schema Design Diagram were implemented programmatically in a script file following best practices. The section below includes the full list of CREATE TABLE statements used to create the EZRental database application.

-- 1 Country

CREATE TABLE Country

```
(
  CountryID      NUMBER(3)      NOT NULL,
  CountryCode2Char CHAR(2)      NOT NULL UNIQUE,
  CountryCode3Char CHAR(3)      NOT NULL UNIQUE,
  CountryName    VARCHAR2(100)  NOT NULL UNIQUE,
  CONSTRAINT pk_Country PRIMARY KEY (CountryID)
);
```

-- 2 USState

CREATE TABLE USState

```
(
  StateID      NUMBER(2)      NOT NULL
    CHECK (StateID BETWEEN 1 AND 75),
  StateCode2Char CHAR(2)      NOT NULL UNIQUE,
  StateName    VARCHAR2(50)   NOT NULL UNIQUE,

  CONSTRAINT pk_USState PRIMARY KEY (StateID)
);
```

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-- 3 DriverLicense

```
CREATE TABLE DriverLicense
(
    DriverLicenseNumber VARCHAR2(25) NOT NULL,
    DriverLicenseExpDate DATE NOT NULL,
    DriverLicenseState CHAR(2) NOT NULL,
    CONSTRAINT pk_DriverLicense PRIMARY KEY (DriverLicenseNumber)
);
```

-- 4 CustomerUserAccount

```
CREATE TABLE CustomerUserAccount
(
    CustomerUserAccountID RAW(16) DEFAULT SYS_GUID() NOT NULL,
    Username VARCHAR2(50) NOT NULL UNIQUE,
    Password VARCHAR2(75) NOT NULL,
    Email VARCHAR2(100) NOT NULL UNIQUE,
    CONSTRAINT pk_CustomerUserAccount PRIMARY KEY (CustomerUserAccountID)
);
```

-- 5 Credit Card Processing Merchant Service Company Table

```
CREATE TABLE CC_ProcessingMerchantSvcCo
(
    CCPMS_CompanyCode NUMBER(2) NOT NULL
        CHECK (CCPMS_CompanyCode BETWEEN 1 AND 20),
    CCPMS_CompanyName VARCHAR2(50) NOT NULL
        UNIQUE,
    CONSTRAINT pk_CCPMS PRIMARY KEY (CCPMS_CompanyCode)
);
```

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-- 6 CreditCardNetworkCompany

CREATE TABLE CreditCardNetworkCompany

```
(
  CreditCardNetworkCompanyCode  NUMBER(2)  NOT NULL
    CHECK (CreditCardNetworkCompanyCode BETWEEN 1 AND 20),
  CreditCardNetworkCompanyName  VARCHAR2(50) NOT NULL
    UNIQUE,
  CONSTRAINT pk_CCN PRIMARY KEY (CreditCardNetworkCompanyCode)
);
```

-- 7 CreditCardIssuingBank

CREATE TABLE CreditCardIssuingBank

```
(
  CreditCardIssuingBankCode  NUMBER(2)  NOT NULL
    CHECK (CreditCardIssuingBankCode BETWEEN 1 AND 20),
  CreditCardIssuingBankName  VARCHAR2(50)  NOT NULL
    UNIQUE,
  CONSTRAINT pk_CCI PRIMARY KEY (CreditCardIssuingBankCode)
);
```

-- 8 CreditCardCorporateMerchantBank

CREATE TABLE CC_CorpMerchantBank

```
(
  CC_CMB_Code  NUMBER(2)  NOT NULL
    CHECK (CC_CMB_Code BETWEEN 1 AND 20),
  CC_CMB_Name  VARCHAR2(50)  NOT NULL
    UNIQUE,
  CONSTRAINT pk_CC_CMB PRIMARY KEY (CC_CMB_Code)
);
```

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-- 9 Discount

CREATE TABLE Discount

```
(
    DiscountID    NUMBER(10)    NOT NULL,
    DiscountCode   CHAR(10)     NOT NULL UNIQUE,
    DiscountCodeDesc VARCHAR2(150) NOT NULL,
    CONSTRAINT pk_Discount PRIMARY KEY (DiscountID)
);
```

-- 10 Company

CREATE TABLE Company

```
(
    CompanyID      NUMBER(5)    NOT NULL,
    CompanyName     VARCHAR2(100) NOT NULL UNIQUE,
    AddressLine1    VARCHAR2(50) NOT NULL,
    AddressLine2    VARCHAR2(50) NULL,
    City            VARCHAR2(50) NOT NULL,
    StateCode       CHAR(2)     NOT NULL,
    Zipcode         VARCHAR2(10) NOT NULL,
    Country         VARCHAR2(100) NOT NULL,
    CompanyRepName  VARCHAR2(50) NOT NULL,
    ContactPhone    VARCHAR2(20) NOT NULL,
    ContactEmail    VARCHAR2(100) NOT NULL UNIQUE,
    CorporateDiscountPercent NUMBER(8,2) NOT NULL,
    CONSTRAINT pk_Company PRIMARY KEY (CompanyID),
    CONSTRAINT ck_CompanyID CHECK (CompanyID BETWEEN 1 AND 20000),
    CONSTRAINT fk_Company_State
        FOREIGN KEY (StateCode)
        REFERENCES USState (StateCode2Char),
    CONSTRAINT fk_Company_Country
        FOREIGN KEY (Country)
        REFERENCES Country (CountryName)
);
```

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-- 11 Customer

```
CREATE TABLE Customer(  
    CustomerID          NUMBER(10)    NOT NULL,  
    FirstName           VARCHAR2(50)   NOT NULL,  
    LastName            VARCHAR2(50)   NOT NULL,  
    BirthDate           DATE           NOT NULL,  
    AddressLine1        VARCHAR2(50)   NOT NULL,  
    AddressLine2        VARCHAR2(50)   NULL,  
    City                VARCHAR2(50)   NOT NULL,  
    StateCode           CHAR(2)        NOT NULL,  
    Zipcode             VARCHAR2(10)   NOT NULL,  
    Country             VARCHAR2(100)  NOT NULL,  
    Phone              VARCHAR2(20)   NOT NULL,  
    Email              VARCHAR2(100)  NOT NULL UNIQUE,  
    DriverLicenseNumber VARCHAR2(25)   NOT NULL,  
    CustomerUserAccountID RAW(16)     NULL,  
    CustomerType        CHAR(1)       NOT NULL  
        CHECK (CustomerType IN ('R','C')),  
    CONSTRAINT pk_Customer PRIMARY KEY (CustomerID),  
    CONSTRAINT fk_Customer_State  
        FOREIGN KEY (StateCode)  
        REFERENCES USState (StateCode2Char),  
    CONSTRAINT fk_Customer_Country  
        FOREIGN KEY (Country)  
        REFERENCES Country (CountryName),  
    CONSTRAINT fk_Customer_DriverLicense  
        FOREIGN KEY (DriverLicenseNumber)  
        REFERENCES DriverLicense (DriverLicenseNumber),  
    CONSTRAINT fk_Customer_UserAccount  
        FOREIGN KEY (CustomerUserAccountID)  
        REFERENCES CustomerUserAccount (CustomerUserAccountID) );
```


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-- 12 CreditCard

CREATE TABLE CreditCard

(

CreditCardNumber VARCHAR2(16) NOT NULL,
CreditCardOwnerName VARCHAR2(100) NOT NULL,
CCPMS_Code NUMBER(2) NOT NULL
CHECK (CCPMS_Code BETWEEN 1 AND 20),
CCN_Code NUMBER(2) NOT NULL
CHECK (CCN_Code BETWEEN 1 AND 20),
CCI_Code NUMBER(2) NOT NULL
CHECK (CCI_Code BETWEEN 1 AND 20),
CCCMB_Code NUMBER(2) NOT NULL
CHECK (CCCMB_Code BETWEEN 1 AND 20),
ExpDate DATE NOT NULL,
AddressLine1 VARCHAR2(50) NOT NULL,
AddressLine2 VARCHAR2(50) NULL,
City VARCHAR2(50) NOT NULL,
StateCode CHAR(2) NOT NULL,
Zipcode VARCHAR2(10) NOT NULL,
Country VARCHAR2(100) NOT NULL,
CreditCardLimit NUMBER(8,2) NOT NULL,
CC_AvailCredit NUMBER(8,2) NOT NULL,
ActivationStatus NUMBER(1) NOT NULL
CHECK (ActivationStatus IN (1,0)),

CONSTRAINT pk_CreditCard PRIMARY KEY (CreditCardNumber),

CONSTRAINT fk_CC_PMS

FOREIGN KEY (CCPMS_Code)

REFERENCES CC_ProcessingMerchantSvcCo (CCPMS_CompanyCode),

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CONSTRAINT fk_CC_Network

FOREIGN KEY (CCN_Code)

REFERENCES CreditCardNetworkCompany (CreditCardNetworkCompanyCode),

CONSTRAINT fk_CC_Issuer

FOREIGN KEY (CCI_Code)

REFERENCES CreditCardIssuingBank (CreditCardIssuingBankCode),

CONSTRAINT fk_CC_CorpBank

FOREIGN KEY (CCCMB_Code)

REFERENCES CC_CorpMerchantBank (CC_CMB_Code)

);

-- 13 Customer_CreditCard

CREATE TABLE Customer_CreditCard

(

CreditCardNumber VARCHAR2(16) NOT NULL,

CustomerID NUMBER(10) NOT NULL,

CONSTRAINT pk_Customer_CreditCard

PRIMARY KEY (CreditCardNumber, CustomerID),

CONSTRAINT fk_CustCard_Card

FOREIGN KEY (CreditCardNumber)

REFERENCES CreditCard (CreditCardNumber),

CONSTRAINT fk_CustCard_Customer

FOREIGN KEY (CustomerID)

REFERENCES Customer (CustomerID));

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-- 14 EZPlus

CREATE TABLE EZPlus

```
(
  EZPlusID          NUMBER(10)  NOT NULL,
  CustomerID        NUMBER(10)  NOT NULL,
  EZPlusRewardsCode CHAR(13)    NOT NULL UNIQUE,
  EZPlusRewardsEarnedPoints NUMBER
    CHECK (EZPlusRewardsEarnedPoints BETWEEN 0 AND 50000),

  CONSTRAINT pk_EZPlus PRIMARY KEY (EZPlusID),

  CONSTRAINT fk_EZPlus_Customer
    FOREIGN KEY (CustomerID)
    REFERENCES Customer (CustomerID)
);
```

-- 15 RetailCustomer

CREATE TABLE RetailCustomer

```
(
  RetailCustomerID NUMBER(10)  NOT NULL,
  CustomerID        NUMBER(10)  NOT NULL,
  EZPlusID          NUMBER(10)  NOT NULL,
  CONSTRAINT pk_RetailCustomer PRIMARY KEY (RetailCustomerID),
  CONSTRAINT fk_RetailCustomer_Customer
    FOREIGN KEY (CustomerID)
    REFERENCES Customer (CustomerID),
  CONSTRAINT fk_RetailCustomer_EZPlus
    FOREIGN KEY (EZPlusID)
    REFERENCES EZPlus (EZPlusID)
);
```

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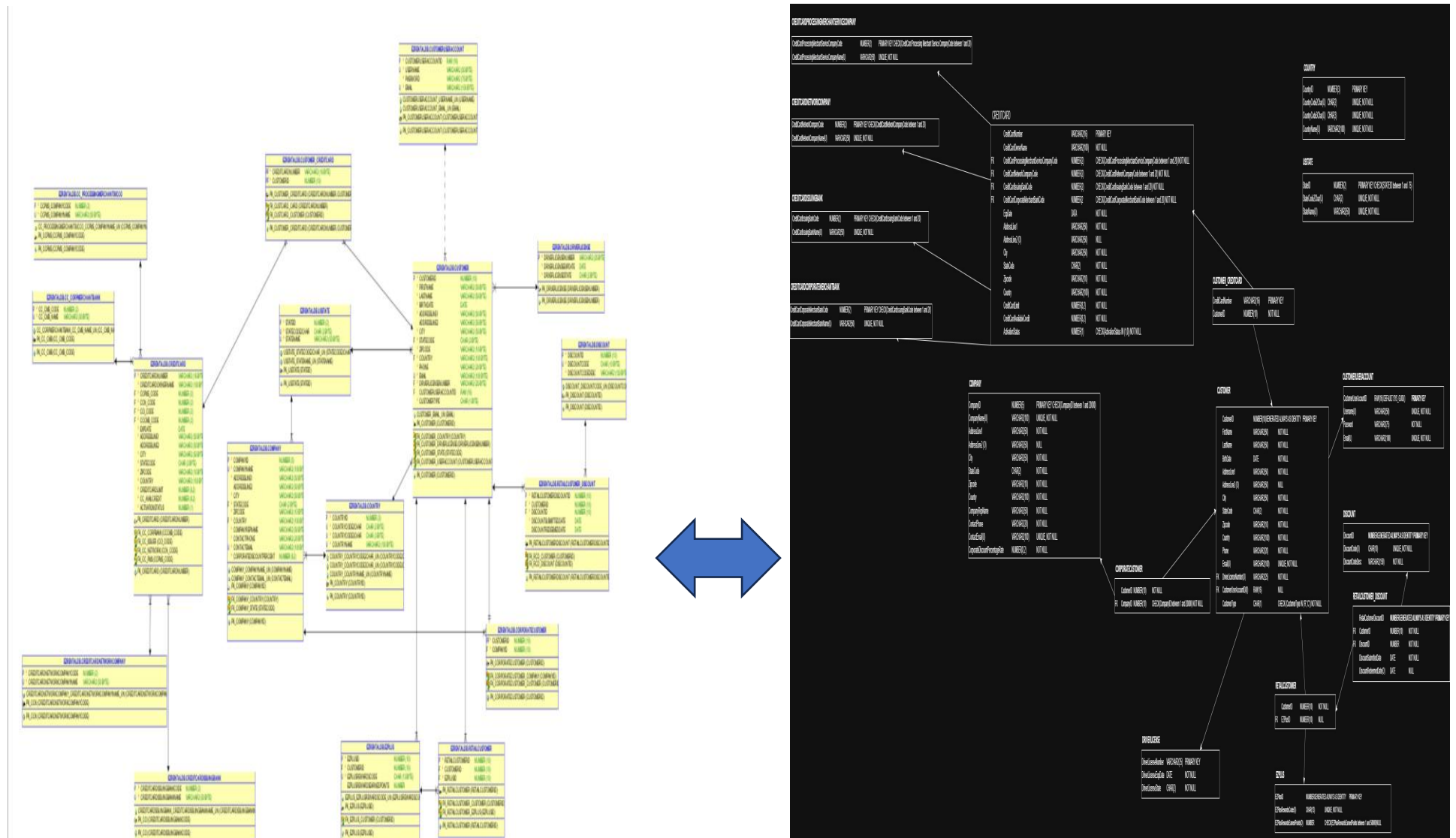
-- 16 RetailCustomer_Discount

```
CREATE TABLE RetailCustomer_Discount      (  
    RetailCustomerDiscountID  NUMBER(10)   NOT NULL,  
    CustomerID                NUMBER(10)   NOT NULL,  
    DiscountID                NUMBER(10)   NOT NULL,  
    DiscountSubmittedDate     DATE         NOT NULL,  
    DiscountRedeemedDate      DATE         NULL,  
    CONSTRAINT pk_RetailCustomerDiscount PRIMARY KEY (RetailCustomerDiscountID),  
    CONSTRAINT fk_RCD_Customer  
        FOREIGN KEY (CustomerID)  
        REFERENCES Customer (CustomerID),  
    CONSTRAINT fk_RCD_Discount  
        FOREIGN KEY (DiscountID)  
        REFERENCES Discount (DiscountID)  
);
```

-- 17 CorporateCustomer

```
CREATE TABLE CorporateCustomer  
(  
    CustomerID  NUMBER(10)  NOT NULL,  
    CompanyID   NUMBER(10)  NOT NULL  
        CHECK (CompanyID BETWEEN 1 AND 20000),  
    CONSTRAINT pk_CorporateCustomer PRIMARY KEY (CustomerID),  
    CONSTRAINT fk_CorporateCustomer_Customer  
        FOREIGN KEY (CustomerID)  
        REFERENCES Customer (CustomerID),  
    CONSTRAINT fk_CorporateCustomer_Company  
        FOREIGN KEY (CompanyID)  
        REFERENCES Company (CompanyID)  
);
```

After implementing all tables and relationships in Oracle Server, the Physical Schema Diagram was generated using Oracle SQL Developer. This diagram shows the exact structure of the EZRental database as implemented and is used to verify that the created schema matches the Physical Schema Design Diagram. The following section includes the finalized implemented diagram.



IMPLEMENTED PHYSICAL SCHEMA DIAGRAM

DATABASE VALIDATION TESTING

CONCLUSION

The EZRental Auto Rental Database Sprint 1 successfully established the foundational analytical and architectural deliverables required for enterprise level system implementation. This included completion of business and technical requirements, the enterprise ER/EER model, multiple normalized logical data models, and structured development planning using Agile methodology. These artifacts provide the critical blueprint for transitioning into physical database design, schema creation, and implementation in Oracle Database during subsequent sprints. With clear data relationships, well defined entities, and a scalable schema, the project is positioned for successful development, testing, and deployment in later phases.